

then this is not the vertex figure of a 4-polytope that is combinatorially equivalent to P^Δ .

(c) 2-Faces of 5-Polytopes Cannot be Prescribed

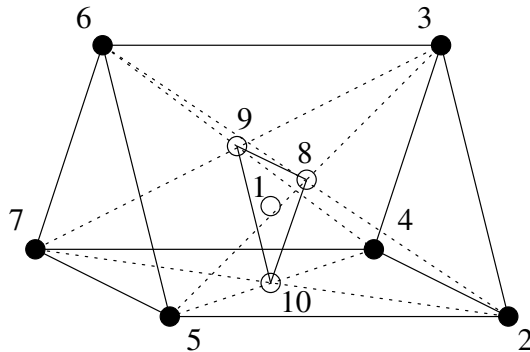
We have now seen the second proof that the shape of a facet cannot be prescribed for 4-polytopes, so we'll have to start and accept it.

But perhaps this was the wrong generalization of the 3-dimensional theorem that a facet can be prescribed. What about prescribing the shape of a 2-face for a d -polytope? The case $d = 4$ was an open problem in the first edition of this book (Problem 6.11*), now you can find it as Exercise 6.11. For $d = 5$ we have the following counterexample from the first edition. It is the type of analysis that Gale diagrams “were made for”: however, the construction of this example may have been the first time that a 3-dimensional affine Gale diagram was seriously used.

Example 6.23. *There is a 5-polytope P with 10 facets and 12 vertices, for which the shape of a 2-face cannot be prescribed.*

To prove this, we construct the polar polytope $Q := P^\Delta$, a 5-polytope with 10 vertices and 12 facets, and verify that for one contraction (face figure, see Exercise 2.9) of a 2-face, the shape cannot be prescribed.

For this consider the signed point configuration given by the vertices of a triangular prism as positive points, labeled $2, 3, \dots, 7$, the centers of the square facets of the prism as negative points $8, 9, 10$, and the center of the whole prism as another negative point 1 . See the figure for how we label this.



It is easy to give coordinates. In fact, the corresponding vector configuration in $\mathbb{R}^4$ could be taken to be

$$\begin{pmatrix} -1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 \\ -1 & 2 & 2 & 2 & 0 & 0 & 0 & -1 & -1 & -1 \\ -2 & 0 & 0 & 6 & 0 & 0 & 6 & 0 & -3 & -3 \\ -2 & 0 & 6 & 0 & 0 & 6 & 0 & -3 & -3 & 0 \end{pmatrix}.$$

Now we check the following facts, which together imply all we need to know.

1. *This signed point configuration is the affine Gale diagram of a 5-polytope Q with 10 vertices.*

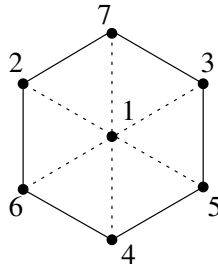
For this we check that every cocircuit of this configuration has at least two negative and two positive elements.

2. *The triple $(8, 9, 10)$ describes a triangle $F = [8, 9, 10]$ which is a 2-face of the 5-polytope Q .*

The point 1 is in the interior of the prism, so the points different from 8, 9, 10 support a positive circuit $(++++++000)$.

3. *The face figure Q/F is a hexagon whose diagonals cross.*

In fact, we get the affine Gale diagram of Q/F by deleting the points 8, 9, 10 from the diagram for Q . But what is left then is the affine Gale diagram of a hexagon $[2, 6, 4, 5, 3, 7]$ with 1 as the intersection point of the long diagonals:



4. *Q is a polytope with 12 facets.*

The facets are the (convex hulls of the points corresponding to the) complements of the positive circuits in the diagram, which are easily enumerated as

286	385	2107	4105	397	496
23157	23167	24156	24167	34156	34157

5. *Every Gale diagram G' with the same positive circuits contains the diagram of the hexagon with crossing diagonals.*

Consider any other (linear) Gale diagram G' on 10 points with the same positive circuits. From the 3-point circuits we see that the sets $2'4'5'7'10'$, $2'3'5'6'8'$, and $3'4'6'7'9'$ have to be planar. However, they cannot collapse to be on a line, because then the whole diagram would collapse to a plane and couldn't have 5-point circuits. From this one can show (using projective uniqueness of the triangular prism — here we are skipping the detailed arguments) that in suitable coordinates $G' \setminus 1'$ coincides with $G \setminus 1$: the Gale diagram G' consists of a triangular prism and its facet centers as well. Now the 5-point circuits imply that the point $1'$ has to be in the interior of this triangular prism. Hence, if we consider the diagram $G' \setminus \{8', 9', 10'\}$, then this has the 5-point circuits listed above (so it describes the right hexagon), and it has 3-point cocircuits $1', 2', 5'$, $1', 3', 6'$, and $1', 4', 7'$ (so the long diagonals of the hexagon cross in $1'$, as required). $\square$

An *explicit* geometric description of the polytope $P = Q^\Delta$ is given in Exercise 6.27.

(d) Polytopes Violating the Isotopy Conjecture

Recall from Section 4.4 that the “realization space” of a polytope is an elementary semialgebraic set, and that elementary semialgebraic sets can be arbitrarily complicated spaces: they can be disconnected, with holes, and so forth (Exercise 4.22).

However, Steinitz’ Theorem 4.11 states that for every 3-dimensional polytope the realization space $\mathcal{R}(P)$ is contractible, and thus connected. (To get this right, we had to fix an affine basis in our Definition 4.10 of realization spaces, to make sure that the “reflection” doesn’t create a second component of the realization space.)

In other words, any two 3-polytopes of the same combinatorial type and orientation can always be continuously deformed into each other, such that each intermediate object is a 3-polytope of the same combinatorial type. The same is not hard to show for d -polytopes with at most $d + 3$ vertices, using Gale diagrams.

However, this “isotopy property” fails even for 4-polytopes: Kleinschmidt constructed a 4-dimensional example with 10 vertices [114]; its combinatorial type is obtained by glueing two copies of the 8-vertex Kleinschmidt polytope of Exercise 6.18 in their octahedron facets in an “incompatible way.” See Mněv [408, p. 530] and Bokowski & Guedes de Oliveira [115]. A systematic construction method for 4-dimensional counterexamples is provided by Richter-Gebert’s Universality Theorem for 4-polytopes [459].

Here we start a construction with a planar point configuration, where the *isotopy conjecture* [463] [234] fails — and transfer this result to polytopes.

Example 6.24 (A nonisotopic 24-polytope with 28 vertices).

The “isotopy property” fails for d -polytopes with $d + 4$ vertices: for that, we can start from a planar point configuration that does not satisfy isotopy. The smallest nonisotopic planar point configurations that are currently known have 14 points. The first such configuration was found by Suvorov [537] [96, p. 363]. Here we present the nicest and newest one, found by Richter-Gebert [458].

For this we refer to the figure on the next page: it represents Richter-Gebert’s example, with two points drawn “at infinity” (which we interpret as “very far away,” to get an affine configuration).

The key property of the figure can be seen as follows. Try to construct a new configuration of 14 points in the (projective) plane, with the condition that point sets that are collinear in the old figure also have to be collinear in the new figure. After a projective transformation, we may assume that the points 1, 2, 3, and 4 (a “projective basis”) are positioned as in the old figure.